

Proof. Theorem (Jensen). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex, i.e.

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad \text{for all } x, y \in \mathbb{R}, t \in [0, 1]. \quad (\text{C})$$

For any integer $n \geq 1$, real numbers $x_1, \dots, x_n$ and non-negative weights $\lambda_1, \dots, \lambda_n$ with $\sum_{i=1}^n \lambda_i = 1$, one has

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (\text{J})$$

Proof by induction on n .

1. Degenerate case. If $\lambda_k = 1$ for some index k and $\lambda_i = 0$ for $i \neq k$, then both sides of (J) equal $f(x_k)$; hence the inequality holds with equality.

2. Base case $n = 2$. Let $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 + \lambda_2 = 1$. Applying (C) with $t = \lambda_1$ gives

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2),$$

which is exactly (J) for two points.

3. Inductive hypothesis. Assume that for some $k \geq 2$ the inequality (J) holds for every collection of k points with non-negative weights summing to 1, i.e.

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) \quad (\text{IH})$$

for all $\lambda_i \geq 0$ with $\sum_{i=1}^k \lambda_i = 1$.

4. Inductive step $k \rightarrow k+1$. Let $\lambda_1, \dots, \lambda_{k+1} \geq 0$ with $\sum_{i=1}^{k+1} \lambda_i = 1$ and set $\mu := \lambda_{k+1}$. If $\mu = 1$ we are in the degenerate situation of Step 1, so assume $\mu \neq 1$.

Define new weights for the first k points

$$\alpha_i := \frac{\lambda_i}{1 - \mu}, \quad i = 1, \dots, k.$$

Because $\sum_{i=1}^k \lambda_i = 1 - \mu$, we have $\alpha_i \geq 0$ and

$$\sum_{i=1}^k \alpha_i = 1.$$

Hence the full convex combination can be written as a two-point mixture:

$$\sum_{i=1}^{k+1} \lambda_i x_i = (1 - \mu) \left(\sum_{i=1}^k \alpha_i x_i \right) + \mu x_{k+1}. \quad (4.1)$$

Applying convexity (??) to the right-hand side of (??) with $t = 1 - \mu$ yields

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq (1 - \mu) f\left(\sum_{i=1}^k \alpha_i x_i\right) + \mu f(x_{k+1}). \quad (4.2)$$

By the inductive hypothesis (??) applied to the inner combination $\sum_{i=1}^k \alpha_i x_i$ we have

$$f\left(\sum_{i=1}^k \alpha_i x_i\right) \leq \sum_{i=1}^k \alpha_i f(x_i). \quad (4.3)$$

Substituting (??) into (??) gives

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq (1 - \mu) \sum_{i=1}^k \alpha_i f(x_i) + \mu f(x_{k+1}). \quad (4.4)$$

Using the definition of α_i and the identity $(1 - \mu)\alpha_i = \lambda_i$, the first term on the right-hand side of (??) simplifies:

$$(1 - \mu) \sum_{i=1}^k \alpha_i f(x_i) = \sum_{i=1}^k \lambda_i f(x_i). \quad (4.5)$$

Consequently,

$$\begin{aligned} f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) &\leq \sum_{i=1}^k \lambda_i f(x_i) + \mu f(x_{k+1}) \\ &= \sum_{i=1}^{k+1} \lambda_i f(x_i), \end{aligned}$$

which is precisely Jensen's inequality for $k + 1$ points.

Thus the statement holds for $k + 1$ whenever it holds for k . Together with the base case $n = 2$ (and the trivial case $n = 1$) the inequality (??) is true for every integer $n \geq 1$.

5. Equality condition. Equality in the final inequality can occur only when equality holds in every step used above.

- In the outer application of convexity (??), equality requires either $\mu \in \{0, 1\}$ (the mixture collapses to a single point) *or*

$$\sum_{i=1}^k \alpha_i x_i = x_{k+1}.$$

- In the inner inequality (??), equality holds precisely when the restriction of f to the convex hull of $\{x_1, \dots, x_k\}$ is affine; for a strictly convex f this forces all those points to be equal (or all but one weight zero).

Hence, in general, equality in Jensen's inequality holds *iff* the points with positive weights lie in a region where f is affine. In particular, if f is *strictly* convex, equality occurs only in the two situations listed in Step1:

1. one weight equals 1 (the degenerate case), or
2. all points $x_1, \dots, x_n$ are equal.

6. Remarks on zero weights. If some $\lambda_i = 0$ the corresponding term disappears from both sides of (??); the argument above is unchanged, so the inequality remains valid. $\square$